

Incremental Gaussian Splatting: Gradual 3D Reconstruction from a Monocular Camera Following Physical World Changes

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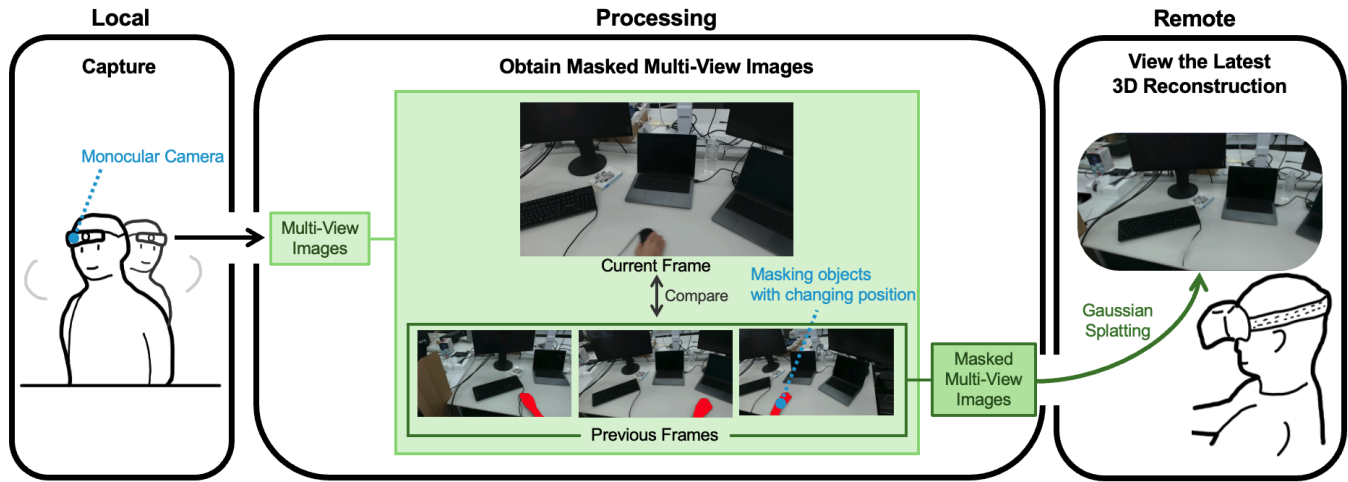


Figure 1: Overview of our Incremental Gaussian Splatting (I-GS) system for real-time 3D reconstruction in dynamic environments. The system captures scenes with a monocular camera, masks moving objects by comparing frames, and applies Gaussian Splatting to masked multi-view images. This enables efficient 3D model updates without accumulating artifacts from dynamic elements, allowing remote users to view the latest reconstruction.

Abstract

Real-time 3D reconstruction is crucial for remote collaboration, but adapting to dynamic environments remains challenging. We introduce Incremental Gaussian Splatting (I-GS), a novel approach for efficient 3D reconstruction in changing scenes. I-GS focuses on human-induced changes, using images from an operator-worn monocular camera. By processing only regions consistent with the current state, I-GS achieves sequential updates. Our evaluation demonstrates that I-GS outperforms conventional Gaussian Splatting, providing accurate spatial representations resilient to moving objects. This advance significantly improves the quality of remote collaboration in dynamic environments.

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1 Introduction

Systems capable of real-time photorealistic 3D reconstruction hold immense potential for revolutionizing remote physical collaboration. In remote work environments, spatial awareness—the ability to perceive relative positions of humans, objects, and surroundings—is crucial, particularly when performing physical tasks from a distance [Komiyama et al. 2017]. Recent advancements in 3D reconstruction techniques have significantly improved the quality and speed of digitizing physical environments. Notably, 3D Gaussian Splatting (3D GS) [Kerbl et al. 2023] has emerged as a faster alternative to Neural Radiance Fields (NeRF), enabling rapid, high-quality 3D reconstructions of physical spaces. This technological progress facilitates swift sharing of 3D-reconstructed workspaces in remote collaborations, enhancing spatial perception for distant participants. However, adapting to dynamically changing physical environments

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remains a significant challenge. Continuous rescanning, necessary to accommodate environmental changes, imposes substantial computational costs and learning time, hindering real-time applications. To address this challenge, we propose Incremental Gaussian Splatting (I-GS) (Fig.1), a novel approach rooted in the observation that most changes in physical environments are induced by human actions. Our method constructs multi-view images from video streams captured by a monocular webcam worn by a human operator. We then perform 3D reconstruction using these multi-view images while excluding regions from previous frames that conflict with the current frame. Specifically, our approach matches feature points between current and past frames, eliminating information inconsistent with the current state before constructing the Gaussian Splatting. Additionally, we mask elements that continuously change position, such as human hands, at the initial stage. This masked multi-view image processing enables high-quality 3D reconstruction of physical environments with limited frames, without accumulating reconstructions of dynamic objects. Consequently, I-GS allows for sequential Gaussian Splatting execution in dynamic environments using a single monocular camera.

2 System Overview

Our system extracts feature points from past and current frames captured by a monocular camera to detect position changes of objects within the images. Objects detected to have changed position are masked, and finally, 3D reconstruction is performed using Gaussian Splatting (GS) (Fig. 2).

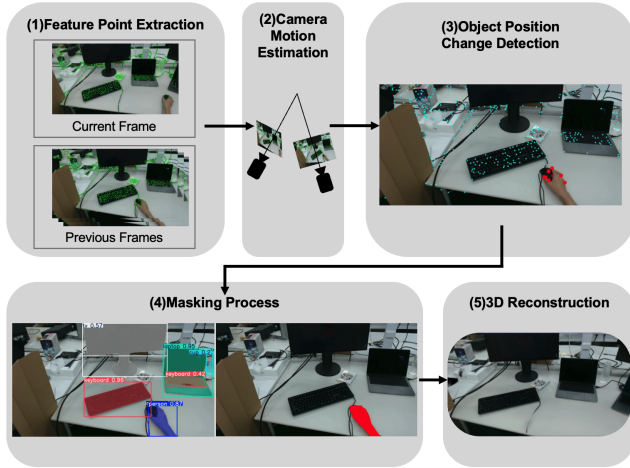


Figure 2: The process consists of five key steps: (1) Feature Point Extraction from current and previous frames, (2) Camera Motion Estimation using the extracted feature points, (3) Object Position Change Detection based on the estimated camera motion, (4) Masking Process to exclude moving objects, and (5) 3D Reconstruction using Gaussian Splatting.

- (1) **Feature Point Extraction:** We rapidly detect feature points from images using the FAST (Features from Accelerated Segment Test) feature detector, and generate feature descriptors around these points using ORB (Oriented FAST and Rotated BRIEF).

- (2) **Camera Motion Estimation:** Feature points are matched between past and current frames. We estimate the monocular camera's motion using homography transformation based on feature point correspondences in the image plane. Furthermore, we use the essential matrix to describe the spatial relationship of corresponding points and precisely estimate changes in camera position and orientation.
- (3) **Object Position Change Detection:** We accurately detect position changes of objects in the physical world by utilizing epipolar constraints obtained from stereo vision generated based on the estimated camera motion.
- (4) **Masking Process:** We apply masks to objects detected to have changed position using the segmentation model of YOLOv8.
- (5) **3D Reconstruction:** Using the masked images as input, we perform 3D reconstruction using GS.

This approach enables efficient and accurate 3D reconstruction in dynamic environments using only a monocular camera.

3 Conclusion and Future Work

Incremental Gaussian Splatting (I-GS) effectively reduces noise by masking outdated information that contradicts the current state of the physical world, surpassing conventional GS methods. As illustrated in Figure 3, I-GS demonstrates robust and accurate 3D reconstruction, even with a limited number of frames, in scenarios where elements change positions. While I-GS shows promising results, we acknowledge certain limitations.



Figure 3: Comparison of conventional Gaussian Splatting (GS) and Incremental Gaussian Splatting (I-GS). The red circles highlight areas where I-GS effectively removes outdated information, resulting in a cleaner 3D reconstruction in dynamic environments.

Acknowledgments

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